Electronic Supplementary Material
Testing past-adapted accounts of anticipatory EEG with post-endpoint randomisation

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This supplement gives the formal and operational detail behind the post-endpoint randomisation test described in the main text. It supports the Level II-A design only. It contains no mechanistic, dynamical, or physical model of any residual; the construction and evaluation of such models are separate, prospectively specified work. Section numbering uses an **S** prefix and is self-contained. All references to "the main text" point to the accompanying Perspective.

Contents

S1 Formal randomisation boundary

This section proves the conditional-exclusion result stated as Proposition 1 in the main text and records the measure-theoretic conditions it needs. The result is elementary; we give it in full because the entire evidential logic of the design rests on it.

S1.1 Setup and filtration

Fix a trial j with randomisation stratum $\mathcal{R}_j$ and committed endpoint window $I_{\text{pre}} = [t_0, t_1]$. Let $(\Omega, \mathcal{A}, \mathbb{P})$ carry the trial process, and let $\{\mathcal{F}_t^{(j)}\}_{t \le t_1}$ be the filtration generated by all pre-assignment information: neural and physiological history, protocol state, elapsed foreperiod, conditional hazard, arousal and vigilance proxies, motor-preparation proxies, block and session state, time on task, and previous-trial variables. The stratum label $\mathcal{R}_j$ is $\mathcal{F}_{t_1}^{(j)}$-measurable. The assigned delay $\tau_L^{(j)} \in [0, T_0]$ is generated after t_1 by an independent service with declared conditional law $P_{\text{sch}}(\mathrm{d}\tau_L \mid \mathcal{R}_j)$.

Definition 1 (Past-adapted statistic)**.** A real-valued statistic $X^{(j)}$ is past-adapted on trial j if it is $\mathcal{F}_{t_1}^{(j)}$-measurable, i.e. computable from information available at or before t_1.

Definition 2 (Exogenous post-endpoint randomisation)**.** The randomisation source is exogenous if, conditional on $\mathcal{R}_j$, the assigned delay is independent of the pre-assignment history,

$$\tau_L^{(j)} \perp\!\!\!\perp \mathcal{F}_{t_1}^{(j)} \mid \mathcal{R}_j. \tag{S1}$$

Equation (S1) is the formal content of condition (R1) under the past-adapted null. It is an assumption about the assignment mechanism, made plausible by concealment and post-endpoint generation and tested by the randomisation, leakage, and implementation-swap audits of Section S7. It is not established by inspecting the assignment software.

S1.2 Conditional exclusion

Proposition S1 (Conditional exclusion of the past-adapted class)**.** *Let $X^{(j)}$ be integrable, $\mathbb{E}|X^{(j)}| < \infty$.*

1. *If $X^{(j)}$ is past-adapted and (S1) holds, then*

$$\mathbb{E}\Big[X^{(j)} \mid \sigma(\tau_L^{(j)}), \mathcal{R}_j\Big] = \mathbb{E}\Big[X^{(j)} \mid \mathcal{R}_j\Big] \quad a.s., \tag{S2}$$

   and for every Borel B with $P_{\text{sch}}(B \mid \mathcal{R}_j) > 0$, $\mathbb{E}[X^{(j)} \mid \tau_L^{(j)} \in B, \mathcal{R}_j] = \mathbb{E}[X^{(j)} \mid \mathcal{R}_j]$.

2. *Equation (S2) applies with $X^{(j)} = A_{\text{pre}}^{(j)}$ and with any prospectively fixed transform $g(A_{\text{pre}}^{(j)})$ whose parameters do not depend on the assignment vector, provided $A_{\text{pre}}^{(j)}$ is past-adapted, which holds under conditions (R1)–(R2).*

3. *Under (R1)–(R3), (S2) extends to the retained analysis sample only if retention, delivery, preprocessing, missingness, and inclusion preserve the randomised contrast; Section S8 makes this condition operational.*

Proof. For part 1, since $X^{(j)}$ is $\mathcal{F}_{t_1}^{(j)}$-measurable and, by (S1), $\tau_L^{(j)}$ is conditionally independent of $\mathcal{F}_{t_1}^{(j)}$ given $\mathcal{R}_j$, the pair $(X^{(j)}, \mathcal{R}_j)$ is conditionally independent of $\tau_L^{(j)}$ given $\mathcal{R}_j$. Hence for any bounded measurable ϕ, $\mathbb{E}[X^{(j)}\phi(\tau_L^{(j)}) \mid \mathcal{R}_j] = \mathbb{E}[X^{(j)} \mid \mathcal{R}_j]\,\mathbb{E}[\phi(\tau_L^{(j)}) \mid \mathcal{R}_j]$. Taking $\phi = \mathbf{1}_B$ and dividing by $P_{\text{sch}}(B \mid \mathcal{R}_j) > 0$ gives the stated set equality; the conditional-expectation form (S2) follows by the defining property of conditional expectation given $\sigma(\tau_L^{(j)}) \vee \mathcal{R}_j$ and a monotone-class argument extending from indicators to integrable $X^{(j)}$. Part 2 is the special case $X^{(j)} = A_{\text{pre}}^{(j)}$ (or $X^{(j)} = g(A_{\text{pre}}^{(j)})$); under (R2) the endpoint and any fixed transform are past-adapted because confirmatory preprocessing is strictly causal and label-blind. Part 3 fails in general once selection acts after assignment: if an inclusion indicator $S^{(j)}$ depends on both $A_{\text{pre}}^{(j)}$ and $\tau_L^{(j)}$, then conditioning on $S^{(j)} = 1$ can induce dependence even when (S2) holds marginally. The magnitude of this induced dependence is bounded in Section S8. □

Table S1: Locked components of the committed pre-event endpoint and comparator.

Component	Locked information
Recording	Modality, sampling rate, channel or source set, montage, reference, online filters (causal only).
Endpoint window	$I_{\text{pre}} = [t_0, t_1]$ relative to the warning cue; baseline interval; closing time t_1.
Feature	Temporal operation (e.g. mean slope or window average), aggregation rule across channels, sign convention.
Eligibility	Trial-inclusion and artefact criteria computable from $\mathcal{F}_{t_1}^{(j)}$; rejection thresholds.
Comparator covariates	Past-adapted covariate vector X_j (Table S2); transforms and interactions.
Folds and loss	Cross-fitting fold structure (block-respecting); comparator family, tuning rule, loss.
Decision constants	Materiality threshold $\beta_{\min}$; $N_{\min}$; randomisation-replicate count; bootstrap settings.

S1.3 Scope of the statement

Proposition S1 is a boundary statement, not a sufficiency theorem for the comparator. It says only that a past-adapted statistic cannot be ordered by a post-endpoint assigned delay under (S1). It does not assert that the forward-only comparator has captured every neural cause of anticipation. Identification of the delay contrast comes from the randomisation mechanism, not from the completeness of any covariate set. A material delay ordering of the committed endpoint therefore localises to one of: (i) failure of (S1) (randomisation or concealment failure); (ii) failure of past-adaptedness of the endpoint or a comparator covariate (temporal leakage, condition (R2)); (iii) a post-randomisation selection pathway (condition (R3)); or (iv) the declared past-adapted account being conditionally insufficient in the tested regime. The audits and sensitivity analysis are built to separate (i)–(iii) from (iv).

S2 Endpoint definition and preprocessing lock

S2.1 Locked endpoint specification

The committed endpoint $A_{\text{pre}}^{(j)}$ is a single-trial, signed, baseline-corrected scalar feature. Before any delay label is accessed, the protocol locks the components in Table S1. Once locked, these are irrevocable for confirmatory analysis; changes define a new regime and require a new protocol.

S2.2 Strict causality and the preprocessing lock

Every confirmatory operation that directly produces the endpoint or a comparator covariate is strictly causal: an output at time $t \le t_1$ depends only on samples at $u \le t$. The following are prohibited for confirmatory endpoint construction: zero-phase, bidirectional, forward–reverse, symmetric, and time-shift-corrected filtering; baseline or detrending steps that use post-t_1 samples; and any preprocessing, rejection, or inclusion rule that depends on the delay label.

Label-blind artefact models are allowed only as nuisance procedures under the same endpoint-lock constraint. Their training data, application window, and rejection rules must be fixed before delay access, and the temporal-leakage audit must show that post-t_1 activity cannot change the committed endpoint beyond the declared tolerance. Any acausal pipeline that fails this requirement may be reported only as an exploratory robustness analysis, clearly separated from the confirmatory result.

S2.3 Temporal-leakage audit

Before delay labels are accessed, the complete locked endpoint-and-comparator pipeline is applied to synthetic audit records that are flat over I_{pre} and carry controlled post-t_1 content: impulses, step

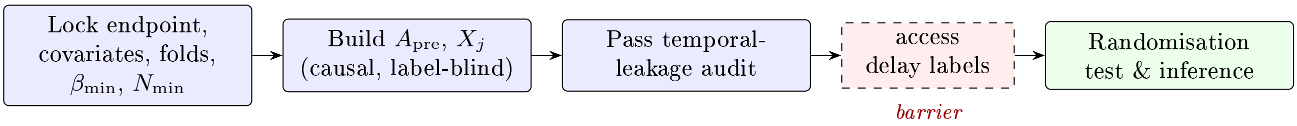

Figure S1: Endpoint and preprocessing lock. All construction occurs label-blind and must pass the temporal-leakage audit before delay labels are accessed.

Table S2: Forward-only comparator covariate class. All entries are past-adapted ($\mathcal{F}_{t_1}^{(j)}$-measurable).

Group	**Covariates**
Timing	Elapsed foreperiod; conditional hazard; scheduler/block probability known pre-t_1.
Slow potential	CNV-like slow-potential amplitude or slope over a pre-t_1 window disjoint from the endpoint feature.
State	Arousal/vigilance proxy (e.g. pupil or alpha-band measure); motor-preparation proxy.
History	Previous-trial timing and outcome; reaction-time history; sequential indices; time on task.
Structure	Participant and session identifiers; preprocessing covariates (e.g. retained-channel counts).
Interactions	Prospectively declared interactions and nonlinear transforms of the above.

responses, and event-locked transients placed at the earliest and latest admissible delivered latencies. The pipeline passes only if these post-endpoint inputs produce no committed pre-event output beyond a declared tolerance δ_{leak}, fixed prospectively as a small fraction of the endpoint's label-blind noise scale. Failure invalidates the confirmatory endpoint irrespective of any later slope. Figure S1 shows the ordering of the lock and the barrier it creates.

S3 Forward-only comparator specification

S3.1 Covariate class

The forward-only comparator absorbs known anticipatory structure. Its covariates are listed in Table S2. Every covariate must be $\mathcal{F}_{t_1}^{(j)}$-measurable. No realised-delay label, and no deterministic transform of one, may enter comparator fitting. Covariates derived from scheduler state are admissible only insofar as they are known before t_1 (e.g. block probabilities, elapsed foreperiod), not the specific post-endpoint draw.

S3.2 Family, cross-fitting and the frozen residual

The comparator family may be high-capacity (penalised regression, gradient boosting, random forests, or recurrent/state-space predictors), with family, tuning rule, fold structure, and loss locked before delay access. Outer cross-fitting folds are fixed prospectively and keep randomisation blocks intact; all hyperparameter selection occurs inside the corresponding training partition. For a trial in fold $k(j)$,

$$A_{\text{pre}}^{\text{resid},(j)} = A_{\text{pre}}^{(j)} - \widehat{m}_0^{(-k(j))}(X_j), \qquad \widehat{m}_0^{(-k)} \approx \mathbb{E}[A_{\text{pre}} \mid X,\ \text{folds} \neq k]. \tag{S3}$$

Held-out predictions and residuals are generated once and frozen before the randomisation test. Refitting or retuning inside the test loop is prohibited.

S3.3 Why residualisation does not identify the contrast

Residualisation improves precision and tests robustness against conventional explanations, but it is not the source of identification. The delay contrast is identified by the assignment mechanism. This matters for validity because a flexible nuisance fit can borrow endpoint information across trials, so the fitted residual $A_{\mathrm{pre}}^{\mathrm{resid},(j)}$ is not focal-trial measurable. The randomisation test therefore holds the entire frozen residual array fixed and regenerates only the assigned-delay vector, under the assignment mechanism actually used. This follows covariate-adjusted randomisation-inference logic[1], extended to cross-fitting compatible with the randomisation design rather than an i.i.d. sampling approximation[2]. Holding all endpoints fixed under reassignment is exact only under a sharp null that also excludes carryover of $\tau_L^{(j)}$ to later endpoints; when carryover is possible the sequential-score route of Section S4 is used.

S4 Confirmatory estimand and inference

S4.1 Participant-level estimand

Within participant p, trial j, stratum $\mathcal{R}_{pj}$, with the assigned delay centred at its scheduler mean $\bar{\tau}_{\mathcal{R}_{pj}} = \mathbb{E}_{P_{\mathrm{sch}}}[\tau_L \mid \mathcal{R}_{pj}]$,

$$A_{\mathrm{pre}}^{\mathrm{resid},(pj)} = \alpha_{\mathcal{R}_{pj}} + \beta_{\tau,p}\big(\tau_L^{(pj)} - \bar{\tau}_{\mathcal{R}_{pj}}\big) + \varepsilon_{pj}, \qquad \beta_\tau = \mathbb{E}_p[\beta_{\tau,p}]. \tag{S4}$$

Each eligible participant contributes one slope, with equal weight. A prospectively fixed rule handles any non-estimable participant identically in every replicate.

S4.2 Intention-to-treat indexing

The primary analysis is indexed by the assigned label $\tau_L^{(j)}$, the quantity protected by randomisation, not by the measured latency $\widetilde{\tau}_L^{(j)} = t_{\mathrm{event}}^{(j)} - t_1$. This preserves the randomisation guarantee under operating-system jitter, display or audio latency, dropped frames, or trigger delay. The measured latency is retained for the delivery audit (Section S7) and for descriptive compliance reporting only.

S4.3 Sequential-score route under carryover

When the delay assigned on trial j may affect the endpoint on later trials, the assignment-isolation sharp null is not exact. The default in that case is the sequential-score route. The current endpoint and residual are treated as part of the pre-assignment history for the current draw, and the participant slope is the estimating-equation coefficient

$$\widehat{\beta}_{\tau,p}^{\mathrm{seq}} = \frac{\sum_{j\in\mathcal{J}_p} A_{\mathrm{pre}}^{\mathrm{resid},(pj)}\,\Delta\tau_{pj}}{\sum_{j\in\mathcal{J}_p} v_{pj}}, \qquad \Delta\tau_{pj} = \tau_L^{(pj)} - \bar{\tau}_{\mathcal{R}_{pj}}, \tag{S5}$$

with $v_{pj} = \mathrm{Var}_{P_{\mathrm{sch}}}(\tau_L \mid \mathcal{R}_{pj})$ the per-trial assignment variance. The centred increment $\Delta\tau_{pj}$ is a martingale difference with respect to the pre-assignment history under (S1), so $\sum_j A_{\mathrm{pre}}^{\mathrm{resid},(pj)}\Delta\tau_{pj}$ has conditional mean zero under the null; the corresponding conditional-randomisation calibration is given in Section S5. The assignment-isolation route is permitted only when predeclared reset or washout diagnostics (lagged endpoint-on-previous-delay checks, pass thresholds, minimum retained-trial yield, and required delay leverage) justify endpoint-array invariance; failure reverts to the sequential-score route or classifies the study as operationally unqualified.

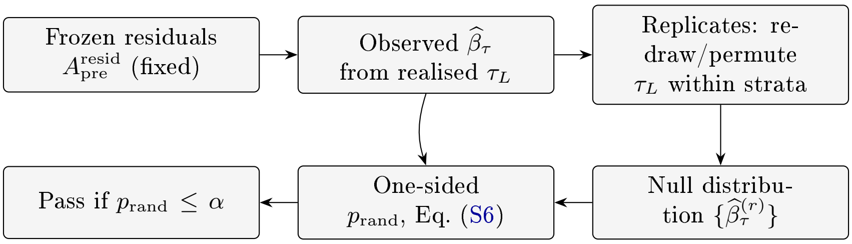

Figure S2: Randomisation-test workflow. The frozen residual array is fixed; only the assigned-delay vector is regenerated under the declared assignment mechanism.

S5 Randomisation test

S5.1 Assignment-isolation route

The test statistic is the equal-participant average slope $\widehat{\beta}_\tau = \frac{1}{P}\sum_p \widehat{\beta}_{\tau,p}$ computed from the frozen residuals. The sharp null states that each committed residual is invariant to admissible reassignment of τ_L. Each replicate reproduces the assignment mechanism actually used: fixed multisets are permuted within their declared blocks, and stochastic draws are redrawn from $P_{\mathrm{sch}}(\mathrm{d}\tau_L \mid \mathcal{R}_{pj})$. No replicate moves an assignment across participant, session, block, or other declared stratum, and the frozen residual array is never recomputed.

S5.2 One-sided p-value

Let $\widehat{\beta}_\tau^{(r)}$, $r = 1, \ldots, R$, be the replicate statistics. The one-sided randomisation p-value, with the finite-sample plus-one correction, is

$$p_{\mathrm{rand}} = \frac{1 + \#\{r : \widehat{\beta}_\tau^{(r)} \leq \widehat{\beta}_\tau\}}{1 + R}. \tag{S6}$$

The number of replicates R is declared prospectively. Where exhaustive enumeration is feasible it replaces the Monte Carlo estimate. The assignment-calibrated criterion passes when $p_{\mathrm{rand}} \leq \alpha$ for the declared one-sided level α[3].

S5.3 Sequential-score calibration

Under the sequential-score route, the statistic $T_p = \sum_j A_{\mathrm{pre}}^{\mathrm{resid},(pj)} \Delta\tau_{pj}$ is calibrated against its conditional-randomisation distribution. Each replicate conditions on the observed pre-assignment histories and redraws the current-trial assigned delays from the declared scheduler within the corresponding strata; the centred contrasts $\Delta\tau_{pj}$ are then recomputed using the same centring rule as in Equation (S5). The residuals are not recomputed. This calibration tests whether the observed martingale-score sum is unusually negative under conditionally random assignment, without assuming that later endpoints would remain unchanged under a counterfactual reassignment of earlier trials.

The observed aggregate statistic

$$T_{\mathrm{seq}} = \frac{\sum_p T_p}{\sum_p \sum_{j \in \mathcal{J}_p} v_{pj}}$$

is compared with the corresponding replicate statistics using (S6), with v_{pj} defined as the scheduler variance in Equation (S5). Figure S2 summarises the workflow.

S5.4 Finite-sample validity and choice of replicate count

The plus-one randomisation p-value of (S6) is exactly valid in finite samples, not only asymptotically. Under the sharp null the observed statistic and the R replicate statistics are exchangeable, so the rank of the observed value among the $R+1$ values is uniform on $\{1,\dots,R+1\}$. Hence

$$\mathbb{P}\Big(p_{\text{rand}} \le \tfrac{k}{R+1}\Big) = \tfrac{k}{R+1}, \quad k = 1,\dots,R+1, \qquad \Longrightarrow \qquad \mathbb{P}(p_{\text{rand}} \le \alpha) \le \alpha \tag{S7}$$

for every $\alpha \in (0,1)$ and every finite R[9]. The $+1$ in numerator and denominator is what makes the test valid rather than anti-conservative; omitting it admits an impossible $p = 0$ and breaks the level. This finite-sample exactness, inherited from the assignment mechanism rather than from a large-sample approximation, is the reason the design uses randomisation inference for the confirmatory test.

Two finite-R considerations then fix the replicate count, declared prospectively. First, granularity: the smallest attainable value is $p_{\min} = 1/(R+1)$, so R must satisfy $p_{\min} < \alpha$ with margin to resolve the decision; with $R = 9{,}999$, $p_{\min} = 1.0 \times 10^{-4}$. Second, Monte Carlo error: p_{rand} estimates the ideal exhaustive-enumeration value $p^\star$ as a binomial proportion, with

$$\widehat{\text{se}}(p_{\text{rand}}) \approx \sqrt{\frac{p^\star\,(1-p^\star)}{R}}. \tag{S8}$$

Near a decision boundary $p^\star \approx \alpha$ the count R is set so this standard error is small relative to α; for $\alpha = 0.05$ and $R = 9{,}999$, $\widehat{\text{se}} \approx 2.2 \times 10^{-3}$, about 4% of α. The same R is used in every calibration, including the $N_{\min}$ and adversarial-null simulations, so that all reported operating characteristics share one Monte Carlo resolution. Where the admissible reassignment set is small enough, exhaustive enumeration removes (S8) entirely and is preferred.

S6 Participant-level bootstrap and materiality threshold

S6.1 Studentised participant bootstrap

The materiality bound uses a studentised participant bootstrap-t. Resample participants with replacement, carrying all of each participant's trials, sessions, strata, and frozen residuals intact; recompute $\widehat{\beta}^*_\tau$ and its participant-level standard error $\widehat{\text{se}}^*$; and form $t^* = (\widehat{\beta}^*_\tau - \widehat{\beta}_\tau)/\widehat{\text{se}}^*$. The one-sided upper confidence bound is

$$\text{UCB}_{0.95}(\widehat{\beta}_\tau) = \widehat{\beta}_\tau - q^*_{0.05}\,\widehat{\text{se}}, \tag{S9}$$

with $q^*_{0.05}$ the 5% quantile of $\{t^*\}$. A bias-corrected and accelerated (BCa) participant bootstrap and a conventional participant t-interval are reported as sensitivity analyses. Trial-level resampling is not used; it would treat within-participant trials as independent replications.

S6.2 Materiality threshold

The materiality criterion is $\text{UCB}_{0.95}(\widehat{\beta}_\tau) < -\beta_{\min}$ for a prospectively declared $\beta_{\min} \ge 0$. Because no validated effect-size scale exists for a delay residual, $\beta_{\min}$ is set to a single-participant resolution floor,

$$\beta_{\min} = \kappa\,\frac{\sigma^{\text{blind}}_{\text{resid}}}{\sigma_\tau \sqrt{\bar{n}_{\text{ret}}}}, \tag{S10}$$

where $\sigma_{\text{resid}}^{\text{blind}}$ is the label-blind within-participant residual noise scale, σ_τ the assigned-delay standard deviation over the declared support, $\bar{n}_{\text{ret}}$ the planned retained trials per participant, and κ a fixed multiple (e.g. $\kappa = 2$) declared before analysis. The within-participant residual scale is used because the participant-slope estimand centres the assigned delay within participant and is invariant to per-participant offsets, so this is the dispersion the slope actually resolves against. All inputs are computed on label-blind qualification data. $\beta_{\min}$ is never selected from the observed $\widehat{\beta}_\tau$, an injected synthetic effect, or an unblinded pilot.

$\beta_{\min}$ is a resolution floor, not a fixed population α-level threshold. The support decision combines the randomisation test, the participant-level upper bound, this floor, the audits, $N_{\min}$, and the selection gate as a non-compensatory intersection. The effective normal-equivalent stringency of the upper-bound component depends on $\beta_{\min}/\text{se}_{\text{pop}}$: under a normal null approximation

$$\Pr\Big\{\text{UCB}_{0.95}(\widehat{\beta}_\tau) < -\beta_{\min}\Big\} \approx \Phi[-\{\beta_{\min}/\text{se}_{\text{pop}} + 1.645\}]. \tag{S11}$$

This approximation varies by many orders of magnitude with se_{pop} and is reported only to interpret the floor; it is not a declared test size, and no claim is made that a particular κ corresponds to a particular one-sided population level. The confirmatory calibration is empirical and design-based, using the randomisation distribution and simulation under the planned operating characteristics, not (S11). Materiality at this stage records only whether a residual is resolvable above noise; whether it is large enough to justify any downstream modelling is a separate question outside this package.

S6.3 Minimum participant count and operating characteristics

The minimum analysable participant count $N_{\min}$ is fixed by simulation under the planned participant heterogeneity, endpoint variance, scheduler, trial count, retention, skewness, and tail behaviour, so that the declared design controls the false-positive decision rate under the sharp and adversarial nulls and recovers the materiality-boundary effect at the planned power. If fewer than $N_{\min}$ eligible participants remain, the assignment-calibrated result may be reported but no population-materiality conclusion is drawn.

S6.4 Finite-sample behaviour of the bootstrap-t bound

The choice of a studentised participant bootstrap-t for $\text{UCB}_{0.95}(\widehat{\beta}_\tau)$ is driven by its finite-sample calibration. Pivoting on the studentised statistic $t = (\widehat{\beta}_\tau - \beta_\tau)/\widehat{\text{se}}$ makes the bootstrap second-order accurate: its one-sided coverage error is $O(P^{-1})$, against $O(P^{-1/2})$ for the normal approximation and for the basic or percentile bootstrap[10;11]. With the participant as the resampling unit and a modest number of participants P, this difference is practical, not cosmetic. The bootstrap-t quantile $q^*_{0.05}$ departs from the normal value -1.645 precisely to the extent that the participant-slope distribution is skewed or heavy-tailed; substituting -1.645 under those conditions would mis-state the bound.

Three finite-sample requirements follow. First, the per-resample standard error $\widehat{\text{se}}^*$ must itself be stable, because it appears in the denominator of every t^*; with small P a large resample count (e.g. $B = 9{,}999$) controls the Monte Carlo error of the 5% quantile. Second, the cluster structure is preserved exactly: participants are resampled with all their trials, sessions, strata, and frozen residuals intact, because trial-level resampling would treat within-participant trials as independent and understate $\widehat{\text{se}}$. Third, a hard floor on P is required: below $N_{\min}$ (Section S6.3) the bootstrap-t quantiles are unstable and no population-materiality claim is made. The bias-corrected and accelerated (BCa) interval,

Table S3: Audit and failure-mode checklist.

Audit	Protects	Pass criterion	Failure classification
Randomisation	(R1)	Assignment balance across strata within tolerance; seed, log, timestamp, and concealment integrity verified.	Invalid test; diagnostic failure.
Temporal leakage	(R2)	Injected post-t_1 impulses/steps produce no committed pre-event output beyond δ_{leak}.	Invalid endpoint; diagnostic failure.
Delivery	(R3)	Assigned-to-measured latency error within declared tolerance; non-compliance bounded per stratum.	Compliance-limited; report bound.
Retention	(R3)	Stratum-specific delivery, rejection, missingness, inclusion recorded; retained-vs-excluded endpoint distributions logged pre-unblinding.	Feeds selection-sensitivity bound.
Implementation swap	(R1)	Association not tied to a particular generator or command pathway across swapped implementations.	Implementation-confounded; diagnostic failure.

whose acceleration is estimated from the participant jackknife, and the ordinary participant t-interval are reported alongside as sensitivity analyses[12]; material disagreement among the three is itself treated as evidence that P is too small for a confident bound. The worked example of Section S9.7 reports all three for one simulated dataset.

S7 Leakage audits and negative controls

S7.1 Audit battery

Four audits, reported regardless of outcome, protect conditions (R1)–(R3). Table S3 lists each audit, the condition it protects, its pass criterion, and the failure classification.

S7.2 Negative controls as two-sided diagnostics

The assigned delay is the primary negative-control exposure: under (R1)–(R3) and a past-adapted account it should show no material association of either sign with the committed endpoint[4;5]. Only a materially *negative* slope is the proposed residual; a material association of either sign on an auxiliary negative control is a design alarm. Recommended auxiliary negative controls include: (i) a pre-cue baseline endpoint computed on a window preceding any anticipatory build-up, which should be unrelated to τ_L; (ii) a future-only "placebo" assignment generated but never delivered, which should not order any endpoint; and (iii) channels or sources outside the locked endpoint set with no anticipatory role. A material association on any auxiliary control of either sign sends the analysis back to the audits before any substantive claim is made.

S7.3 Opposite-direction departures

A material positive slope is reported separately as an opposite-direction negative-control departure. It does not support the proposed residual, and it is not dismissed as a simple null: it triggers the same audit review, implementation checks, and selection-sensitivity analysis. If it survives them, it is, by Proposition S1, conditional insufficiency of the past-adapted account in the unpredicted direction, on the same evidential footing as a surviving negative slope and distinguished only by its inconsistency with the pre-registered prior. Attributing it to artefact on the basis of its sign alone would contradict the sign-blind audits that adjudicate all departures.

S8 Selection and omitted-pathway sensitivity analysis

S8.1 The post-randomisation selection problem

Condition (R3) cannot be closed by inspection. Even under perfect randomisation, an unmeasured pre-event state $U^{(j)}$ becomes a threat if it affects both the committed endpoint and a delay-dependent selection event $S^{(j)}$ (delivery compliance, artefact rejection, missingness, or inclusion). Conditioning the analysis on $S^{(j)} = 1$ can then induce an association between $A_{\text{pre}}^{(j)}$ and $\tau_L^{(j)}$ that mimics a residual. The sensitivity analysis bounds how large such a selection pathway would have to be to manufacture the observed slope, and compares it with the imbalance the audits permit.

S8.2 Required versus audited imbalance

Let Δ_{sel} denote a scalar summary of differential, endpoint-correlated retention or inclusion imbalance across the delay support (e.g. the difference in retained-endpoint means per unit delay attributable to selection). Two quantities are computed:

- the *required* imbalance $\Delta_{\text{sel}}^{\text{req}}$: the smallest Δ_{sel} that reproduces the observed $\widehat{\beta}_\tau$ under an otherwise past-adapted account, mapped through a declared selection-model class;
- the *audited* imbalance $\Delta_{\text{sel}}^{\text{aud}}$: the largest selection imbalance consistent with the stratum-specific delivery, retention, preprocessing, inclusion, and exclusion records, including retained-against-excluded endpoint distributions available before unblinding.

The mapping from imbalance to induced slope uses monotone sharp bounds under a monotone-selection assumption[6] and worst-case bounds without monotonicity[7], in the sensitivity tradition for observational threats[8].

A minimal implementable audit compares retained and excluded trials within each declared stratum. The protocol first computes, without using the outcome decision, how different the committed endpoint is between retained and excluded trials. It then computes how unevenly retention varies across assigned-delay labels within the same stratum. The product of these two quantities, aggregated with the declared analysis weights, gives a conservative estimate of how much endpoint shift could be induced by delay-dependent retention. This audited selection bound is compared with the slope required to explain the observed assigned-delay ordering. More conservative monotone or worst-case bounds may be used instead, but the mapping from retention imbalance to possible endpoint shift must be declared before unblinding and reported together with the confirmatory result.

S8.3 Confidence-limit gate

The selection gate compares confidence limits, not point estimates, so sampling uncertainty in both quantities enters the decision:

$$\mathrm{LCB}_{0.95}\big(\Delta_{\text{sel}}^{\text{req}}\big) > \mathrm{UCB}_{0.95}\Big(\Delta_{\text{sel}}^{\text{aud}}\Big). \tag{S12}$$

If (S12) holds, the observed slope is larger than any selection pathway the audits permit, and the selection gate passes. If the slope is material but (S12) fails, the result is reported as selection-limited, not as support. The selection-model class, the imbalance summary, and the bound estimators are locked prospectively.

S8.4 Scope boundary: endpoint-by-delay collider selection

The gate (S12) is built on the *marginal* retention imbalance across delay bins and on a declared selection-model class. It therefore protects only against selection pathways represented by that audited imbalance summary and that class. Marginal retention balance is not sufficient to exclude selection. Consider a pure endpoint-by-delay collider in which the inclusion indicator S_{pj} depends on the joint configuration of the committed endpoint A_{pj} and the assigned delay $\tau_L^{(pj)}$ through their interaction, with no main effect of delay on retention:

$$\operatorname{logit}\Pr(S_{pj}=1)=c_0+\gamma\,\tilde{A}_{pj}\,\tilde{\tau}_L^{(pj)}, \tag{S13}$$

where $\tilde{A}$ and $\tilde{\tau}_L$ are standardised and centred. Because the inclusion probability is symmetric in $\tilde{A}$ at each delay, the marginal retention rate by delay bin shifts only at second order in $\gamma\tilde{\tau}_L$ and remains approximately balanced, while the retained endpoint distribution shifts linearly across delay bins, manufacturing a retained-sample slope. The full pre-selection sample still satisfies the post-endpoint randomisation boundary of §1: A_{pj} is independent of $\tau_L^{(pj)}$ before selection. The induced association arises only after conditioning on the retained analysis sample, so this is a scope test of the selection gate, not a failure of the randomisation boundary.

Because the committed endpoint is computed for every trial before the delay is assigned, the collider is detectable directly, independently of the marginal imbalance summary. Two predeclared diagnostics are used. The first is an endpoint-by-delay interaction test for inclusion: a logistic regression of S_{pj} on $\tilde{A}_{pj}$, $\tilde{\tau}_L^{(pj)}$, and their product over all assigned trials, with the interaction coefficient tested by its Wald statistic. The second compares the committed endpoint of retained and excluded trials within each delay bin by a Welch statistic, with multiplicity control across bins. A genuine endpoint-level residual does not trip either diagnostic, because under random retention the retained and excluded endpoints share a distribution within a delay bin; a collider trips both, because inclusion depends on the endpoint within off-centre bins. A retained-sample endpoint-rank-by-delay statistic is reported as a descriptive quantity only, because a genuine injected residual would also shift retained endpoints across delay bins and trip it.

The decision rule adds a hard support-blocking condition. If the endpoint-by-delay interaction diagnostic fires, or if the retained-sample slope is compatible with an uncovered collider of the observed magnitude under approximately balanced marginal retention, the dataset cannot receive supported-residual classification; it is classified selection-limited, R3-unqualified, or operationally unqualified. The manufactured collider slope is bounded by the retention rate, so at a high resolution floor and high retention it may remain sub-material and be blocked incidentally by materiality; at the realistic artefact-rejection rates and resolution floors where it becomes material, the scalar gate (S12) passes and only this diagnostic prevents a supported classification (§9).

S8.5 Worked selection-sensitivity calculation

We make the gate (S12) concrete with explicit Lee and Manski numbers, using the locked anchor of Section S9.3 ($T_0 = 20\,\mathrm{ms}$, support $[0, 20]\,\mathrm{ms}$) and the supported dataset of Section S9.7, in which the locked analysis returns $\widehat{\beta}_\tau = -60.6\,\mu\mathrm{V\,s^{-1}}$ with within-participant residual scale $\sigma_y = 1.09\,\mu\mathrm{V}$. The slope corresponds to a retained-endpoint mean difference between the extreme delay bins of $|\widehat{\beta}_\tau|\,T_0 = 60.6 \times 0.020 = 1.21\,\mu\mathrm{V}$. The question is whether differential, endpoint-correlated retention across the delay support could manufacture a shift of that size. Overall retention is $\bar{p} = 0.80$; for the bounded-support arguments the endpoint is Winsorised at the label-blind $\pm 3\sigma_y$ limits.

Manski worst-case bound. Without a monotonicity assumption, and using only the overall retention rate, each bin's pre-selection mean is bounded within $\pm(1-\bar{p})\,3\sigma_y = \pm 0.65\,\mu\text{V}$ of the retained mean, so the worst-case between-bin difference is $1.31\,\mu\text{V}$, an induced slope up to $65\,\mu\text{V}\,\text{s}^{-1}$[7]. This exceeds the observed $60.6\,\mu\text{V}\,\text{s}^{-1}$: the unconstrained worst case, which ignores the realised retention pattern and lets the missing mass sit adversarially at the Winsorising limits, does not by itself exclude a selection explanation. That is the expected behaviour of a worst-case bound, and it is why the gate relies on the audited retention pattern rather than on $\bar{p}$ alone.

Audited differential retention. Because the delay is randomised after the endpoint closes, retention should not depend on it, and the retention audit measures the realised dependence. In the worked dataset the audit returns retention $p_0 = 0.808$ in the shortest-delay bin and $p_1 = 0.792$ in the longest, a differential $\Delta_{\text{sel}}^{\text{aud}} = 0.016$ with one-sided upper bound $\text{UCB}_{0.95}(\Delta_{\text{sel}}^{\text{aud}}) \approx 0.06$.

Lee monotone bound at the audited imbalance. Under monotone selection, the induced between-bin shift is bounded by trimming the more-retained bin to the retention of the less-retained bin, a fraction $t = (p_0 - p_1)/p_0 = 0.016/0.808 = 0.0198$. For a Gaussian residual the resulting one-sided mean ambiguity is $\sigma_y\,\phi(z_t)/(1-t) = 1.0 \times 0.0480/0.980 = 0.049\,\mu\text{V}$, with $z_t = \Phi^{-1}(t) = -2.06$[6]. The induced slope is therefore at most $0.049/0.020 = 2.5\,\mu\text{V}\,\text{s}^{-1}$ (with sampling uncertainty, $\text{UCB}_{0.95} \approx 3.9\,\mu\text{V}\,\text{s}^{-1}$): under monotone, audited selection at most about 4% of the observed slope is attributable to retention imbalance.

Required imbalance and the gate. Inverting the Lee map, manufacturing the full $1.21\,\mu\text{V}$ shift requires trimming a fraction t^{req} of one bin solving the inverse-Mills relation $\phi(c)/(1-\Phi(c)) = 1.21/\sigma_y$, giving $c \approx 0.46$, kept fraction $1 - \Phi(c) \approx 0.32$, hence $t^{\text{req}} \approx 0.68$ and a required differential retention $\Delta_{\text{sel}}^{\text{req}} \approx 0.54$ (one-sided lower bound $\text{LCB}_{0.95} \approx 0.50$). The gate (S12) then compares $\text{LCB}_{0.95}(\Delta_{\text{sel}}^{\text{req}}) \approx 0.50$ with $\text{UCB}_{0.95}(\Delta_{\text{sel}}^{\text{aud}}) \approx 0.06$: the slope would need about a 54-percentage-point differential retention to arise from selection, while the audit bounds it near two points, so the gate passes with a wide margin. The worked dataset is therefore not selection-limited under either the monotone or the audited-retention reading. The Manski paragraph records the honest caveat that an adversarial, non-monotone selection pattern uncoupled from the audited retention record cannot be excluded by retention rates alone, which is exactly why the monotonicity class and the audited imbalance summary are declared prospectively and reported in full. It is also why the marginal-imbalance gate is supplemented by the endpoint-by-delay collider diagnostic of Section S8.4: a balanced collider leaves $\Delta_{\text{sel}}^{\text{aud}}$ near zero and would pass this gate, and is caught instead by the interaction diagnostic.

S9 Synthetic benchmark design

S9.1 Role and the seven required behaviours

The benchmarks verify pipeline behaviour and are not evidence about human EEG. Before any delay labels are analysed, the locked pipeline is run on simulated data and must satisfy seven behaviours:

1. **Forward-only null.** Under a past-adapted data-generating process with no delay dependence, the randomisation test rejects at its nominal rate and $\text{UCB}_{0.95}(\widehat{\beta}_\tau)$ does not clear $-\beta_{\min}$.

2. **Injected residual.** Under an endpoint-level injected residual, the pipeline recovers the negative sign and approximate magnitude with the planned estimators and sample sizes.

3. **Leakage detection.** Under an injected post-t_1 leak, the temporal-leakage audit flags it and support is blocked.

4. **Standard selection.** Under a delay-dependent retention artefact with marginal imbalance, the retention audit fires and the selection gate (S12) moves as the imbalance is varied.

5. **Endpoint-by-delay collider.** Under a pure collider with approximately balanced marginal retention, the dataset is classified selection-limited rather than supported, caught by the endpoint-by-delay diagnostic and not by the scalar gate (Section S8.4).

6. **Adversarial forward-only null.** Under nonlinear hazard structure, heavy-tailed participant heterogeneity, heteroskedastic and autocorrelated residuals, carryover, and comparator misspecification, false-support control holds.

7. **Opposite-direction injection.** Under a positive injected slope, the result is classified as an opposite-direction departure, not as directional support.

A pipeline that fails any behaviour is reported as operationally unqualified rather than as evidence for or against the sufficiency claim. The realised operating characteristics are tabulated in Section S9.6 from the public benchmark package; the worked example in Section S9.7 uses a single simulated dataset to show the decision rule step by step.

S9.2 Data-generating processes

The forward-only generator builds the endpoint from the comparator covariates plus past-adapted noise, with no term in τ_L. The injected-residual generator adds a term $\beta^{\text{inj}}(\tau_L - \bar{\tau}_{\mathcal{R}})$ at the endpoint-generating level, before comparator fitting and residualisation. The leakage generator adds a post-t_1, delay-correlated component to the committed endpoint and to a pre-endpoint probe, used to confirm the temporal-leakage audit fires. The standard selection generator imposes a delay-dependent inclusion rule with marginal imbalance; the collider generator imposes a pure endpoint-by-delay inclusion rule with approximately balanced marginal retention (Section S8.4). The adversarial forward-only generator adds nonlinear hazard structure, heavy-tailed participant heterogeneity, heteroskedastic and autocorrelated residuals, carryover from the previous assigned delay, and a comparator the linear model cannot fully absorb. Each generator is run across the planning grid below; participant heterogeneity is introduced by drawing participant slopes and noise scales from declared distributions.

S9.3 Synthetic anchor and planning grid

The reference benchmark uses a short-horizon common support $T_0 = 20\,\text{ms}$ with five assigned-delay bins $\tau_L \in \{0, 5, 10, 15, 20\}\,\text{ms}$ and, where a residual is injected, an additive slope at the endpoint-generating level. Table S4 reports planning power for the injected-materiality-boundary effect as a function of retained trials per bin and the relative residual-noise setting r_ε. With moderate noise the benchmark reaches 80% power by about five retained trials per bin ($N_{\text{ret}} \approx 25$); under the more conservative $r_\varepsilon = 0.50$ case it requires about twenty retained trials per bin ($N_{\text{ret}} \approx 100$). These are synthetic retained-sample requirements, not participant-level EEG guarantees. An empirical protocol must translate them into a participant-by-trial factorisation using label-blind estimates of residual

Table S4: Planning power for the injected materiality-boundary effect at the synthetic anchor ($T_0 = 20\,\mathrm{ms}$, five bins). Values are illustrative planning targets from the locked Monte Carlo benchmark, not empirical results.

Retained trials/bin n_{rep}	N_{ret}	Power ($r_\varepsilon = 0.30$)	Power ($r_\varepsilon = 0.50$)
5	25	≈ 0.80	≈ 0.45
10	50	≈ 0.95	≈ 0.63
20	100	> 0.99	≈ 0.80

Table S5: Benchmark scenarios and locked design constants (run `0a775ac4f1ead9d3`). The collider scenario uses a lower retention rate and $\kappa = 1$ so the manufactured slope is material and isolates the endpoint-by-delay diagnostic; all other scenarios use $\kappa = 2$ at 80% retention.

Scenario	Generator	trials/bin	retention	$\beta_{\min}$	β^{inj}
Anchor (clean null)	forward-only	12	0.80	40.5	—
Injected residual	endpoint-injected	12	0.80	44.2	-60
Leakage	forward-only + leak	12	0.80	47.4	—
Standard selection	monotone selection	12	0.80	45.2	—
Collider selection	endpoint-by-delay collider	20	0.45	20.6	—
Adversarial null	adversarial forward-only	12	0.78	47.2	—
Opposite direction	endpoint-injected	12	0.80	44.3	$+60$

variance, participant heterogeneity, retained-trial yield, delivery error, and the declared materiality threshold.

S9.4 Support as a design variable

The admissible delay support trades leverage against comparator burden. A wider support increases the assigned-delay variance σ_τ^2 and the slope leverage, but lengthens the interval over which ordinary foreperiod and hazard structure operate, enlarging the burden on the comparator. A narrower support cleans the contrast but drives leverage toward zero. The support is fixed in the label-blind qualification sample at the point balancing these effects and is not revised using delay information.

S9.5 Reporting per benchmark

For each generator and grid point, report: the declared T_0 and bin grid; the realised false-positive decision rate under the null; the recovery rate and sign for the injected residual; audit-firing rates for leakage and selection generators; and the selection-gate behaviour across imbalance levels. Report one-sided 95% bootstrap bounds as the decision objects, with two-sided intervals as descriptive summaries only.

S9.6 Realised operating characteristics

Tables S5 and S6 report the realised operating characteristics of the locked pipeline on the seven scenarios, produced by the public benchmark package (run hash `0a775ac4f1ead9d3`, $M = 1000$ Monte Carlo datasets per scenario, $P = 24$ participants, support $[0, 20]\,\mathrm{ms}$, residual scale $1\,\mu\mathrm{V}$). Every value is generated by the executable code and recorded in the run metadata; no entry is set by hand. The complete machine-generated table with all columns is archived in the repository at `outputs/<run_hash>/tables/`.

The clean anchor holds false support at 0/1000 with the randomisation test rejecting at 0.050, and the adversarial forward-only null also holds false support at 0/1000 under nonlinearity, heavy

Table S6: Realised outcome rates over $M = 1000$ datasets per scenario (run `0a775ac4f1ead9d3`). Slopes and bounds in $\mu\mathrm{V\,s^{-1}}$. "rand" is the randomisation-test pass rate, "resol" the resolution-floor (materiality) pass rate, "leak" and "reten" the leakage and retention audit-fire rates, "gate" the scalar selection-gate pass rate, "coll" the endpoint-by-delay collider-diagnostic fire rate. The gate is decision-relevant only when a material slope is present; for the pure nulls (anchor, adversarial) no material slope arises and the entry is marked —.

Scenario	$\overline{\widehat{\beta}}_\tau$	med. UCB	rand	resol	leak	reten	gate	coll	supp	sel-lim	d-fail	null
Anchor	-0.1	$+7.0$	0.050	0.000	0.001	0.001	—	0.007	0.000	0.000	0.001	0.999
Injected	-59.9	-51.7	1.000	0.935	0.002	0.002	1.000	0.005	**0.928**	0.005	0.002	0.065
Leakage	-85.0	-78.0	1.000	1.000	**1.000**	0.000	1.000	0.003	0.000	0.000	1.000	0.000
Selection	-38.4	-30.2	1.000	0.002	0.003	**1.000**	0.997	1.000	0.000	0.002	0.003	0.995
Collider	-35.1	-28.0	1.000	0.982	0.003	0.004	**0.981**	**1.000**	0.000	**0.979**	0.003	0.018
Adversarial	$+0.0$	$+8.3$	0.051	0.000	0.000	0.000	—	0.004	0.000	0.000	0.000	1.000
Opposite	$+60.3$	$+68.7$	0.000	0.000	0.002	0.000	1.000	0.009	0.000	0.000	0.002	0.045

Table S7: Collider-selection scope sweep (simulated data). The manufactured retained-sample slope grows with collider strength while the marginal retention imbalance stays small; the endpoint-by-delay interaction diagnostic fires throughout.

γ	med. $\widehat{\beta}_\tau$	marg. imbal.	reten. fire	resol. pass	interaction fire	sel.-lim.
-0.500	-17.4	0.030	0	0.005	1.0	0.005
-0.900	-25.7	0.029	0.005	0.260	1.0	0.260
-1.4	-31.5	0.032	0	0.790	1.0	0.790
-2.0	-35.0	0.036	0.005	0.980	1.0	0.980
-2.6	-36.8	0.037	0.005	0.995	1.0	0.995
-3.2	-38.1	0.038	0.005	1.0	1.0	1.0

tails, autocorrelation, heteroskedasticity, carryover, and misspecification. The endpoint-level injected residual is supported in 92.8% of datasets. Leakage and standard selection are blocked by their audits and never supported. The opposite-direction injection is classified as an opposite-direction departure in 95.3% and never as support.

The collider row is the scope test. The pure endpoint-by-delay collider manufactures a material retained-sample slope ($\overline{\widehat{\beta}}_\tau = -35.1$, resolution pass 0.982) with approximately balanced marginal retention (audit-fire 0.004). The scalar selection gate passes in 98.1% of datasets, that is, it does not flag the collider; the endpoint-by-delay diagnostic fires in all datasets and blocks support, so the scenario is classified selection-limited in 97.9% and supported in 0%. Table S7 sweeps the collider strength: the manufactured slope grows from -17 to $-38\,\mu\mathrm{V\,s^{-1}}$ while the marginal retention imbalance stays near 0.03 and never fires the retention audit, the scalar gate passes throughout, and the interaction diagnostic fires throughout. This is the empirical content of the scope boundary: marginal retention balance is insufficient to exclude collider selection, and the scalar gate protects only within its declared marginal-imbalance and selection-model class.

S9.7 Worked numeric example: one dataset through every decision object

This subsection walks a single simulated dataset through every object in the decision rule, to show how the pieces combine. The numbers are illustrative outputs of the locked pipeline on simulated data; they are not empirical results.

Table S8: Retained residual-endpoint means by delay bin (supported draw, replicate 946). The bin-mean standard error is about $\sigma^{\text{blind}}_{\text{resid}}/\sqrt{229} = 0.07\,\mu\text{V}$.

Assigned delay τ_L (ms)	0	5	10	15	20
Centred $\Delta\tau$ (ms)	−10	−5	0	+5	+10
Mean $A^{\text{resid}}_{\text{pre}}$ (μV)	+0.57	+0.35	−0.01	−0.34	−0.59

Table S9: The twenty-four participant slopes $\widehat{\beta}_{\tau,p}$ ($\mu\text{V}\,\text{s}^{-1}$, supported draw, replicate 946). Equal-participant mean $\widehat{\beta}_\tau = -60.6$; standard deviation 20.5; bootstrap standard error 4.2.

−47	−57	−69	−61	−72	−94	−62	−66
−81	−40	−48	−105	−42	−25	−48	−73
−81	−76	−46	−39	−23	−51	−71	−77

Design and locked constants. The dataset has $P = 24$ participants, each contributing about 48 retained trials (twelve per bin before retention) under 80% retention, at the anchor grid $\tau_L \in \{0, 5, 10, 15, 20\}$ ms, so the centred delay has standard deviation $\sigma_\tau = 7.07 \times 10^{-3}$ s. The forward-only residual has within-participant standard deviation close to $\sigma^{\text{blind}}_{\text{resid}} = 1.09\,\mu\text{V}$ after the comparator. From (S10) with $\kappa = 2$ the materiality floor is

$$\beta_{\min} = \kappa\,\frac{\sigma^{\text{blind}}_{\text{resid}}}{\sigma_\tau\sqrt{\bar{n}_{\text{ret}}}} = \frac{2 \times 1.09}{7.07 \times 10^{-3} \times \sqrt{47.7}} = 44.7\,\mu\text{V}\,\text{s}^{-1},$$

which equals two single-participant slope standard errors ($\sigma^{\text{blind}}_{\text{resid}}/(\sigma_\tau\sqrt{\bar{n}_{\text{ret}}}) = 22.3\,\mu\text{V}\,\text{s}^{-1}$). The pre-declared inference constants are $\alpha = 0.05$, $R = 9{,}999$ randomisation replicates, $B = 9{,}999$ participant bootstrap resamples, and $N_{\min} = 10$. This dataset is replicate 946 of the injected-residual scenario in the frozen run `0a775ac4f1ead9d3`.

Supported draw. Data are generated by the injected-residual process with population mean slope $-60\,\mu\text{V}\,\text{s}^{-1}$ injected at the endpoint-generating level and between-participant slope dispersion $12\,\mu\text{V}\,\text{s}^{-1}$. The forward-only comparator is fitted and the residuals frozen (Section S3.2). The retained residual bin means trace the injected slope (Table S8), and the twenty-four participant slopes (Table S9) average to $\widehat{\beta}_\tau = -60.6\,\mu\text{V}\,\text{s}^{-1}$ with a participant-slope standard deviation of $20.5\,\mu\text{V}\,\text{s}^{-1}$, giving a participant bootstrap standard error of $4.2\,\mu\text{V}\,\text{s}^{-1}$.

The randomisation test holds these residuals fixed and regenerates the assigned delays $R = 9{,}999$ times under the scheduler. The observed average slope is more negative than every replicate, so by (S6) the one-sided p-value is at the granularity floor, $p_{\text{rand}} = (1 + 0)/(1 + 9{,}999) = 1.0 \times 10^{-4} \le \alpha$ (Section S5.4). The studentised participant bootstrap-t (Section S6.1) returns a 5% quantile $q^*_{0.05} = -1.68$, slightly beyond the normal -1.645 because the participant-slope distribution is mildly left-skewed, so by (S9)

$$\text{UCB}_{0.95}(\widehat{\beta}_\tau) = \widehat{\beta}_\tau - q^*_{0.05}\,\widehat{\text{se}} = -60.6 - (-1.68)(4.2) = -53.6\,\mu\text{V}\,\text{s}^{-1}.$$

The BCa interval (−53.9) and the participant t-interval (−53.4) agree to within $1\,\mu\text{V}\,\text{s}^{-1}$, so the bound is stable (Section S6.4). Since $-53.6 < -\beta_{\min} = -44.7$, the materiality criterion passes with a margin of $8.9\,\mu\text{V}\,\text{s}^{-1}$. The analysable count $N = 24 \ge N_{\min} = 10$. The audits pass: the temporal-leakage probe shows a maximum delay correlation of 0.002 against a critical value of 0.097; assignment balance is within tolerance; the 99th-percentile delivery error is 0.5 ms against a 2 ms tolerance; and the

Table S10: Every decision object for the two simulated draws through the locked pipeline. Units $\mu\mathrm{V\,s^{-1}}$ for slopes and bounds.

Decision object	**Criterion**	**Supported draw**	**Null draw**
Estimate $\widehat{\beta}_\tau$	—	-60.6	-0.1
Randomisation p ($R = 9{,}999$)	≤ 0.05	1.0×10^{-4} ✓	0.56 ✗
Bootstrap-t $\mathrm{UCB}_{0.95}$	$< -\beta_{\min} = -44.7$	-53.6 ✓	$+9.7$ ✗
Analysable N	$\ge N_{\min} = 10$	24 ✓	24 ✓
Temporal-leakage correlation	$< r_{\mathrm{crit}} = 0.10$	0.002 ✓	0.017 ✓
Delivery error (99th pct)	$< 2\,\mathrm{ms}$	0.5 ms ✓	0.5 ms ✓
Retention differential	within tolerance	1.5 pp ✓	3.0 pp ✓
Collider interaction z	does not fire	0.7 ✓	0.6 ✓
Selection gate $\mathrm{LCB}_{0.95}(\Delta^{\mathrm{req}}_{\mathrm{sel}}) > \mathrm{UCB}_{0.95}(\Delta^{\mathrm{aud}}_{\mathrm{sel}})$	gate holds	$0.50 > 0.06$ ✓	not triggered
Decision	all hold	**Supported residual**	**Forward-only adequate**

retention differential across the delay support is 1.5 percentage points. The endpoint-by-delay collider diagnostic does not fire (interaction Wald $z = 0.7$). The selection-sensitivity gate (Section S8.5) holds: $\mathrm{LCB}_{0.95}(\Delta^{\mathrm{req}}_{\mathrm{sel}}) \approx 0.50 > \mathrm{UCB}_{0.95}(\Delta^{\mathrm{aud}}_{\mathrm{sel}}) \approx 0.06$. All conditions of the non-compensatory rule hold, so the dataset receives a *supported residual* classification; programme-level support still requires independent replication.

Null draw and the contrast. Regenerating from the forward-only process (population mean slope zero, same comparator and noise; replicate 514 of the anchor scenario) gives $\widehat{\beta}_\tau = -0.1\,\mu\mathrm{V\,s^{-1}}$, $p_{\mathrm{rand}} = 0.56$, and $\mathrm{UCB}_{0.95}(\widehat{\beta}_\tau) = +9.7\,\mu\mathrm{V\,s^{-1}}$, which does not clear $-\beta_{\min}$; the audits still pass and no material slope is present for the selection gate to address. The dataset is classified *forward-only adequate*. Table S10 places every decision object side by side for the two draws, demonstrating that the same locked pipeline supports a residual only when one is injected and returns the null otherwise.

S10 Reproducibility checklist

1. **Locked before unblinding.** Endpoint specification (Table S1); comparator covariates and family (Table S2); cross-fitting folds; materiality threshold $\beta_{\min}$ and its formula (S10); $N_{\min}$; randomisation-replicate count R and level α; bootstrap settings; selection-model class and gate (S12).
2. **Randomisation record.** Scheduler $P_{\mathrm{sch}}(\cdot \mid \mathcal{R})$; stratum structure; seeds; logs; timestamps; concealment mechanism; implementation-swap plan.
3. **Pipeline integrity.** Strictly causal preprocessing; temporal-leakage audit result with tolerance δ_{leak}; frozen residual array (S3); no refitting inside the test loop.
4. **Delivery and retention.** Assigned-vs-measured latency record; stratum-specific delivery, rejection, missingness, inclusion; retained-vs-excluded endpoint distributions logged pre-unblinding.
5. **Inference.** Route (assignment-isolation or sequential-score) with its diagnostics; randomisation p-value (S6); bootstrap $\mathrm{UCB}_{0.95}$ (S9); BCa and t-interval sensitivity analyses.
6. **Decision.** The non-compensatory rule (the resolution floor, the randomisation test, the participant-

level bound, $N_{\min}$, the audits, the selection-sensitivity gate, and the endpoint-by-delay collider diagnostic) and the resulting outcome class; explicit classification of any audit failure as invalid rather than null.

7. **Benchmarks.** Generator code and the planning grid (Table S4); realised null decision rate; injected-residual recovery; audit-firing rates; sensitivity-gate calibration; the endpoint-by-delay collider scope test and sweep (Tables S6 and S7).

8. **Replication.** For programme-level support, an independent sample analysed with the identical locked endpoint, comparator family, materiality threshold, inference, and audit logic.

9. **Code and archive.** The executable benchmark package (source, configuration files, environment files, run scripts, tests, generated tables, figure source data, and run metadata) at `https://github.com/asopasakis/cri-leveliia-benchmark`, archived at Zenodo under DOI `10.5281/zenodo.PLACEHOLDER`. All benchmark numbers in this article were generated from the frozen output directory with run hash `0a775ac4f1ead9d3` ($M = 1000$) and can be reproduced with `python scripts/run_all.py --all` and `python scripts/verify_outputs.py`.

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